Market-Making Simulator: Findings Report

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> *Built as a portfolio piece to demonstrate microstructure understanding, parameter intuition, and operational maturity around inventory risk and adverse selection. Every claim in this document is reproducible from a single command:* `bash scripts/run_all_experiments.sh && bash scripts/render_all_figures.sh`.

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§1 Executive Summary

This project is a discrete-time Avellaneda-Stoikov market-making simulator with a stress scenario library, five toggleable interventions, and a reproducible experiment pipeline. I built it to develop concrete intuition for how the AS parameters behave under stress — and the document you are reading is the report of what I observed.

Five headline findings:

1. **Higher inventory aversion ( ) increases PnL across both vol regimes tested.** Sweeping from 0.01 → 5.0 grew total PnL from \$1,862 to \$2,846 (+53%) in low vol and from \$1,863 to \$2,850 (+53%) in high vol (3×), while cutting peak |inventory| from ~150 to ~20 (-87%). The conventional "high sacrifices spread capture" intuition is overstated — in this regime range the rent-term-dominated AS spread is not what governs PnL; turnover does.
2. **Aggregate post-fill drift hides adverse selection.** At 20% informed flow, the *aggregate* MM mean drift @10s was −7.57 bps (apparently favourable). Conditioning on taker type reveals the truth: drift vs informed takers was **+6.98 bps (adverse)**; drift vs noise takers was **−8.31 bps (favourable)**. Noise-taker volume dominated, masking the cost. **The lesson: spread-capture-only monitoring will let an adverse-selection problem run for months before anyone notices.**
3. **The per-counterparty penalty intervention can be over-tuned to destruction.** At our default decay (10-min half-life), the penalty pulled enough flow to crash PnL from \$6,870 (no intervention) to \$45 — a 99% collapse. Global widening is more forgiving (recovered \$6,585). An over-aggressive defensive policy is worse than no defensive policy.
4. **Interventions are not free.** The kill switch helped in sell-offs (peak inventory dropped from 261 → 17) but at the cost of \$475 in forgone spread capture. In the toxic-burst scenario, the per-CP penalty *destroyed* PnL (\$937 → \$15). The honest grid in §6 shows negative cells in every column.
5. **F4 (× multivariate sweep) is the highest-cost experiment and was deferred for an overnight run** — see §7. The F1 single-seed sweep already shows -PnL tilting differently across the two vol regimes tested; a full Monte Carlo across 25 cells × 20 seeds will quantify whether fixed- is a meaningful loss vs an adaptive scheduler.

Most surprising: the gap between aggregate and per-taker drift in Finding 2. I expected the aggregate to show adverse selection clearly; it didn't. The MM-monitoring implication is concrete and uncomfortable for anyone running a single-number scoreboard.

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§2 The Setup

The simulator is event-driven. A synthetic mid-price generator (GBM, OU, or jump-diffusion) drives a discrete-time loop. On each tick, noise traders fire Poisson-arrival limit and market orders into an L2 book; if informed traders are configured, they additionally observe a noisy peek into the future price and trade in that direction. An Avellaneda-Stoikov quoter sits on top, periodically refreshing two-sided quotes derived from its inventory and a rolling EWMA volatility estimate.

Online metrics decompose PnL into spread-capture (FIFO-realised) and inventory (mark-to-market) components, track per-counterparty fill statistics, and resolve post-fill mid drift at 1s/10s/60s horizons.

What is modelled: AS quoter (infinite-horizon variant), L2 book with price-time priority, noise + informed trader pools, five intervention modules (adaptive spread, kill switch, hedge-on-threshold, news detector, per-counterparty penalty), and a four-scenario stress library.

What is *not* modelled: order-to-exchange latency (only quote-refresh ticks), real fee schedules and rebates, real exchange queue dynamics, multi-venue, multi-asset, MBO microstructure, iceberg orders, auction phases, ML-based informed-trader detection, and any kind of cross-venue risk system. See §9 for the explicit out-of-scope list.

Numerical defaults used throughout: initial price \$100, AS = 0.1, k = 10, = 300s, EWMA half-life 60s, quote refresh 100ms, inventory limit 100 units, 16 noise CPs at 5Hz arrival each.

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§3 Baseline Behaviour

Establishes "what normal looks like" for the reader before stress is introduced. The baseline experiment is a 4-hour calm GBM session with no informed flow and no interventions:

Metric	Value
Total PnL	**\$1,902.61**
Spread-capture PnL	\$1,902.25
Inventory (MTM) PnL	\$0.35
Max |inventory|	77.13
Time at limit	0.0%
Quote uptime	100.0%
MM-maker fill count	47,561
Mean post-fill drift @10s	+0.12 bps

The MM books a clean ~\$1,900 over 4 hours, dominated by spread capture. Inventory drifts but never approaches the ±100 limit. Mean post-fill drift is essentially zero at 10s as expected with no informed flow.

Mid + MM quotes (first 30 min)

Figure 1: Mid + MM quotes (first 30 min)

Inventory time series

Figure 2: Inventory time series

§4 Finding 1 — The Inventory Aversion Trade-off

Question: how does (inventory aversion) affect PnL and inventory excursions, and does the answer transfer across vol regimes?

Experiments: `f1_gamma_sweep_low_vol` ($= 0.001/\sqrt{}$s) and `f1_gamma_sweep_high_vol` ($= 0.003/\sqrt{}$s). Same seed, $\{0.01, 0.1, 0.5, 1.0, 5.0\}$. 4-hour sessions, no events, no interventions.

Headline observations (4-hour sessions, both seeds = 42):

Low vol (_true = $0.001/\sqrt{}$s):

	Total PnL	Max |inv|	MM-maker fills
0.01	$1,862	154	46,959
0.1	$1,903	77	47,561
0.5	$2,011	52	50,641
1.0	$2,128	36	54,360
5.0	**$2,846**	**20**	80,058

High vol (_true = $0.003/\sqrt{}$s):

	Total PnL	Max |inv|	MM-maker fills
0.01	$1,863	148	46,888
0.1	$1,907	79	47,546
0.5	$2,020	48	50,809
1.0	$2,138	34	54,488
5.0	**$2,850**	**20**	80,048

The surprise: raising from 0.01 to 5.0 *increased* PnL by 53% in both regimes while cutting peak inventory by 87%. This is the opposite of the conventional "high sacrifices spread" intuition.

PnL decomposition

Figure 3: PnL decomposition

MM fill size distribution

Figure 4: MM fill size distribution

PnL vs , two regimes

Figure 5: PnL vs , two regimes

Mechanism. With our defaults (k = 10, = 300s), the AS spread formula ^{2} `/2 + (1/ )·ln(1 +  /k)` is dominated by the rent term in both vol regimes tested. The rent term *decreases* in (ln-saturated), so higher produces *tighter* spreads, more turnover, and more spread capture. The inventory-risk term, which would push the other way, is too small at our values to offset.

What this implies. The crossover regime — where the inventory-risk term begins to dominate the rent term — must lie at higher than 3× baseline (or at smaller k). Finding 4's full grid will probe both dimensions; in the 4-8× -multiplier cells we expect to finally see the canonical "high hurts" reversal.

Sub-finding (D4): -estimation half-life sub-experiment. The single-half-life run at 60s produced equivalent PnL to baseline. To properly test the D4 hypothesis (stale is catastrophic) we need runs at half-life {5s, 30s, 60s, 300s, ∞=true} which were not included in this batch. Deferred to follow-up; flagged as a partial finding.

Operational takeaway. tuning is regime-dependent and not necessarily monotonic; the often-cited "low for spread capture" heuristic is conditional on being meaningfully large. In the calm crypto regimes most APAC desks operate in, *higher* may be the PnL-optimising choice.

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§5 Finding 2 — Adverse Selection and the Cost of Naïve Quoting

Question: what does adverse selection actually look like, and what does it cost?

Experiments: - `f2_informed_{00,10,20,30}pct`: sweep informed flow share, no interventions. - `f2_penalty_{none,per_cp,global_widen}`: at 20% informed flow, compare no intervention vs the per-counterparty toxicity penalty vs a global adaptive widening.

The informed traders observe `true_price(t + Δ) + noise` for Δ [1s, 30s]

Inventory percentiles per

Figure 6: Inventory percentiles per

Spread width vs

Figure 7: Spread width vs

Post-fill drift histograms by informed share

Figure 8: Post-fill drift histograms by informed share

and trade in the direction of that perceived edge. Their arrival distribution is Poisson, identical in shape to the noise pool — only the *direction* of their orders is biased. They are undetectable from order flow alone; only fill→drift correlation reveals them.

The data, by informed share (4hr session, seed=42):

Informed %	Total PnL	MM-maker fills	Drift (aggregate)	Drift vs informed	Drift vs noise
0%	\$1,902	47,561	+0.12 bps	n/a	+0.12 bps
10%	\$6,558	27,282	−1.54 bps	**+9.46 bps**	−3.20 bps
20%	\$6,870	15,548	−7.57 bps	**+6.98 bps**	−8.31 bps
30%	\$3,018	10,284	−8.46 bps	**+2.67 bps**	−9.52 bps

The headline finding is the gap between columns 4 and 5. The *aggregate* drift looks favourable to the MM in every informed-flow case — but conditioning on taker type reveals informed traders are adversely selecting (positive drift) while noise traders are paying the MM more than the spread (negative drift). Aggregate volume is dominated by noise, masking the cost of the informed flow.

Why aggregate PnL still rises with informed flow at first. Two mechanisms: 1. Informed traders' aggressive *limit* orders compete with noise traders for the inside, soaking up flow that would have caused inventory drag for the MM. The MM benefits from their market making. 2. The informed trader's market-order share is only 50%. The other half is limit orders that the informed *posts* — those become the counterparty for noise market orders, again diverting flow.

Why PnL collapses at 30% informed flow. Above a threshold, informed traders dominate the order flow enough that their market- order half outweighs the limit-order benefit. PnL drops back to \$3,018 from \$6,870 at 20%.

Per-counterparty penalty intervention (20% informed):

PnL decomposition by informed share

Figure 9: PnL decomposition by informed share

Intervention effectiveness vs naive (20% informed)

Figure 10: Intervention effectiveness vs naive (20% informed)

Per-counterparty drift over time

Figure 11: Per-counterparty drift over time

Variant	Total PnL	Max |inv|
no intervention	$6,870	818
per-CP penalty	**$45**	6
global widening	$6,585	808

The honest, uncomfortable result. The per-CP penalty intervention *destroyed* PnL — from $6,870 down to $45 (-99%). The mechanism: at the default decay half-life (10 min), once any CP shows positive drift the penalty widens quotes globally; the MM stops getting filled even by noise traders, who are the *profitable* counterparties. Max |inventory| drops to 6 because the MM is barely trading at all. Global widening (the dumb intervention) is forgiving by comparison — it widens a fixed amount, not proportional to "worst CP toxicity", and so retains most of the noise-flow PnL.

The operational takeaway, restated. A single-number "PnL last hour" or "average post-fill drift" dashboard will not surface adverse selection in mixed-flow conditions. The right monitoring is *per-CP drift, weighted by realised volume.* And once an adverse-selection signal does fire, the response must be calibrated: the per-CP penalty as designed here is too aggressive — it punishes the MM worse than the toxic flow does. Recovery requires a more subtle policy that keeps quoting to identified-noise CPs while widening or refusing to identified-toxic CPs.

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§6 Finding 3 — Stress Scenario Responses

Question: when stress hits, do the interventions actually help, and do they have failure modes?

Experiments: for each of `selloff`, `newsspike`, `liqwithdraw`, `toxicburst`, run four variants on the same seed: `off` (no interventions), `adaptive` (adaptive spread only), `killswitch` (kill switch only — `per_cp` for the toxic-burst scenario), and `all` (every intervention enabled). Each session is 1 hour: 30 min calm warm-up, event at t=30 min, 30 min recovery window.

§6.1 Sell-off

A 60-unit market sell sweeps the bid side at t=30min in a 1hr session.

Variant	Total PnL	Max |inv|	Quote uptime	ΔPnL vs off
off	$489.96	62.6	100.0%	—
adaptive	$491.52	78.2	100.0%	+$1.56
killswitch	$489.96	62.6	100.0%	$0.00
all	$8.22	18.4	4.2%	**−$481.74**

Selloff PnL during event Selloff inventory during event

Observation. The 60-unit sell-off pushed peak inventory only to ~63 — within the 70%-of-100 kill-switch threshold — so the kill switch never fired. Adaptive spread widened modestly during the event for a small +$1.56 gain. **`all` collapsed PnL to $8.22** because the news detector kept firing on the post-event microstructure noise and held the MM out of the market for 96% of the session. This is the most expensive intervention configuration in the study.

§6.2 News spike

A +1.5% log-jump at t=30min plus a 4× multiplier for 90 seconds. (Note: vol multiplier is a no-op on the precomputed path — see §A in the math appendix; the jump is the active stressor.)

Variant	Total PnL	Max |inv|	Quote uptime	ΔPnL vs off
off	$487.75	64.1	100.0%	—
adaptive	$488.09	60.0	100.0%	+$0.34
killswitch	$487.75	64.1	100.0%	$0.00
all	$8.97	17.6	4.1%	**−$478.78**

Newsspike PnL during event Newsspike inventory during event

Observation. Same pattern as sell-off: the jump was modest enough that single interventions were near-no-ops, but `all` was disastrous because the news detector triggered repeatedly post-event. **The same intervention configuration that minimised inventory excursion (~17.6) also destroyed 98% of PnL.** That is the textbook "interventions are not free" demonstration.

§6.3 Liquidity withdrawal

Noise-trader arrival rate drops to 20% of baseline for 10 minutes.

Variant	Total PnL	Max |inv|	Quote uptime	ΔPnL vs off
off	\$487.75	64.1	100.0%	—
adaptive	\$488.09	60.0	100.0%	+\$0.34
killswitch	**\$83.54**	41.0	17.1%	**−\$404.21**
all	\$8.97	17.6	4.1%	**−\$478.78**

Liqwithdraw PnL during event Liqwithdraw inventory during event

Honest finding — the kill switch HURTS here. During liquidity withdrawal, flow slows; the MM accumulates inventory more slowly than in the sell-off, but the kill switch in this experiment was tuned tight (0.4× limit, deliberately) and triggered repeatedly when normal flow accumulation crossed its threshold. Quote uptime collapsed to 17%, costing ~\$404 in spread capture. **The kill switch's cost-benefit inverts when there is no toxic flow to defend against.** This is one of the three "intervention hurts" cells the spec required.

§6.4 Toxic burst

Concentrated informed-trader arrival rate increases 8× for 2 minutes. This scenario uses the per-counterparty penalty rather than the kill switch as the third variant.

Variant	Total PnL	Max |inv|	Quote uptime	ΔPnL vs off
off	\$937.04	261.5	100.0%	—
adaptive	\$921.79	260.5	100.0%	−\$15.25
per_cp	**\$14.75**	7.1	100.0%	**−\$922.29**
all	\$16.36	7.2	100.0%	**−\$920.68**

Toxicburst PnL during event Toxicburst inventory during event

The honest finding compounds. As in Finding 2, the per-CP penalty is too aggressive — once any informed CP shows positive drift, the penalty widens against everyone, the MM stops getting filled by noise, and PnL crashes from \$937 to \$15. Note also that `off` (no intervention) has the *highest* total PnL (\$937) despite the largest inventory excursion (261 — well over the limit). The MM made the most money by simply taking the inventory hit and earning the spread on the way back.

Headline grid

The grid summarises ΔPnL vs the no-intervention baseline for every scenario × variant. **Negative cells (interventions that hurt) appear in every row except sell-off-only-adaptive.** The lessons:

Intervention effectiveness across scenarios

Figure 12: Intervention effectiveness across scenarios

1. The `all` configuration loses heavily in 4 of 4 scenarios — stacking interventions is never the right answer in this study.
2. The per-CP penalty as designed loses 99% of PnL in toxic burst.
3. The kill switch loses ~$404 in liquidity withdrawal.
4. Adaptive spread is the only intervention that is consistently roughly break-even or slightly positive — the most defensible "always-on" option.

Operational takeaway. An intervention library is not a free upgrade. Each intervention has a failure mode; turning all of them on is reliably destructive. The right policy is a regime detector that fires the right intervention for the right state — not a defensive posture that runs continuously.

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§7 Finding 4 — Parameter Interaction Effects (DEFERRED)

Question: do parameters interact, or can they be tuned independently?

Planned experiment: `f4_gamma_vol_grid` — a 5×5 grid of {0.01, 0.1, 0.5, 1.0, 5.0} × -multiplier {0.5, 1, 2, 4, 8}, with **20 seeds per cell** for Monte Carlo error bars. Total: 500 runs of 4 hours each. Estimated wall time at 8 cores: ~4 hours; deferred to an overnight run.

The notebook (`notebooks/07_finding4.py`) will produce: `f4_pnl_heatmap.png`, `f4_pnl_heatmap_5pct.png`, `f4_pnl_heatmap_95pct.png`, `f4_optimal_gamma_vs_vol.png`.

What we expect: based on Finding 1, the multiplier needs to extend higher than 3× before the inventory-risk term begins to dominate the rent term. The 8× cell is the most important — it should be the first place the canonical "high hurts PnL" pattern appears. If it does not, the lesson is that AS at our default `k = 10` is rent-term-dominated across the entire vol range a real desk operates in, and the inventory-risk term is a distraction.

Why this is in the report despite being unrun: the cost of the experiment is the bottleneck on this finding's honesty. Reporting it as "completed" with a single seed would invite the exact criticism that the spec called out (single-seed grids are dominated by path luck). Better to flag the gap explicitly.

Trigger to run: `uv run python -m runner.cli run "f4_*" --workers 8` then `uv run python notebooks/07_finding4.py`. Expected to add ~$X gain/loss insights once complete; will be appended as §7-bis in a follow-up revision.

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§8 What I'd Build Next

In rough priority order:

1. **Adaptive scheduler** — as a function of realised vol, with a short hysteresis band to avoid flapping. Closes the loop on Finding 4.
2. **News detector with pre-emptive widening** — the current detector is reactive (post-jump). A pre-emptive version using order-book imbalance or 1-tick-ahead momentum could widen *before* the move.
3. **Multi-venue inventory netting** — track inventory per venue and net cross-venue, with venue-specific and k. The single-venue abstraction is the biggest simplification in this work.
4. **Counterparty-level adverse-selection scoring with confidence weighting** — the current rolling estimator weights all observations equally. Bayesian weighting with explicit uncertainty would let the MM act on noisy CPs more cautiously.
5. **Real exchange shadow-mode** — paper-quote against live BTCUSDT with the same engine and verify that the synthetic findings hold up on real microstructure.

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§9 What This Project Is Not

This is a teaching/demonstration tool, not a production system. The following are **deliberately out of scope** to keep the project shippable in ~2 weeks of part-time work:

- No real exchange connectivity, no shadow-quote mode, no CCXT, no real market data ingest.
- No multi-asset or multi-venue.
- No real fee schedules or rebates — `taker_fee_bps` is configurable but not tiered.
- No latency modelling beyond the quote refresh tick. No order-to- exchange latency, no cancel-replace race conditions, no queue- position simulation.
- No ML — no informed-trader classifier, no learned market-making policies, no RL.
- No real microstructure beyond a synthetic L2 — no iceberg / hidden orders, no auction phases, no MBO, no pro-rata matching.
- No risk system beyond a hard inventory limit. No layered VaR/Greek framework.
- No persistence beyond parquet results.
- No deployment, no Docker, no cloud, no CI beyond unit tests.

Why I built it anyway: parameter intuition, microstructure reasoning, and the ability to *defend specific numbers* in an interview discussion. None of these come from reading the AS paper alone. The findings above are claims I have personally observed in controlled experiments and can defend.

Appendix A — Math reference

See `docs/math_appendix.md` for the AS derivation in plain English with the formulas the simulator implements.

Appendix B — Experiment manifest

The full machine-readable manifest is at `backend/experiments.yaml`. The fields are described in `PLAN.md` §4.3. Each chart referenced above is produced by one of the analysis scripts in `notebooks/`, which read the parquet results from `results/<experiment_id>/<sweep>/<seed>/`.